

McDonald's AI Drive-Thru Failure (2023–2024)

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2026-03-11

Introduction

McDonald's is one of the most recognizable brands on the planet, serving around 69 million customers every day. When the company decided to test artificial intelligence (AI) at its drive-thrus, it seemed like a natural fit — fast food and fast technology. But what followed was a very public and sometimes comical failure that offers important lessons about implementing new technology in customer-facing environments.

This case asks a fundamental question that any organization must answer before deploying AI: **Just because we can automate something, should we?**

Background: The Drive-Thru Problem

Drive-thrus are the backbone of McDonald's business. In the United States, they account for roughly **70% of McDonald's sales**. They are also a major operational challenge. Customers want speed. Staff face pressure, noise, accents, children shouting orders from the back seat, and constant menu changes. Getting an order right while keeping the line moving is genuinely difficult.

McDonald's leadership saw AI voice ordering as a potential solution. The vision was appealing: an AI system that could take orders consistently, never get tired, never mishear a customer the same way twice (in theory), and potentially upsell items intelligently.

The Partnership with IBM

In 2021, McDonald's acquired a startup called Apprento, which had developed AI voice-ordering technology. McDonald's then sold that technology to **IBM** in 2021 and entered into a partnership to deploy it at drive-thrus. The system, called **Automated Order Taking (AOT)**, used voice recognition and natural language processing to understand what customers were saying and translate it into a point-of-sale order.

Natural language processing (NLP) is a branch of AI that helps computers understand human speech and text — the same technology behind Siri, Alexa, and Google Assistant. It sounds simple, but human language is enormously complex. People say the same thing in dozens of different ways. They have accents. They pause mid-sentence. They change their minds. They speak over each other.

McDonald's began piloting AOT at about **100 locations** across the United States, primarily in Chicago, to test the system in real-world conditions.

What Went Wrong?

The problems became very apparent — and very public. Customers began posting videos on social media, particularly TikTok, showing the AI system making absurd errors:

- A customer ordering **butter** received an order of **69 packets of butter** added to their meal
- A customer trying to order a **simple ice cream cone** had multiple unrelated items added to their order after the AI misunderstood repeated attempts to correct it
- Customers ordering specific items found themselves with elaborate, incorrect orders worth far more than they intended
- The AI frequently struggled with accents, background noise, and non-standard menu phrasing
- When the AI was confused, it would sometimes default to adding items rather than asking for clarification — creating large, wrong orders

In many cases, a human employee had to step in to correct the order anyway — defeating the purpose of the automation. Reports indicated that the system was only completing orders without human intervention a fraction of the time.

In **June 2024**, McDonald’s announced it was ending its partnership with IBM and discontinuing the AOT program at all locations. A spokesperson said the company would “evaluate [its] options” for AI drive-thru technology in the future, but the experiment was effectively over.

Why Did It Fail? An MIS Perspective

1. The Gap Between Lab and Reality

AI systems are developed and tested in controlled environments. In a lab, engineers can use clear recordings, standard phrasing, and predictable inputs. A real McDonald’s drive-thru is the opposite: loud, chaotic, and unpredictable.

This is a common problem in technology deployment called the **production gap** — the difference between how well a system works in testing versus how well it works in the real world. The drive-thru environment proved far more challenging than the system could handle.

2. Voice Recognition Is Hard

Even the best commercial voice recognition systems (from Google, Amazon, Apple) struggle with accents, background noise, and unusual phrasings. McDonald’s menu is not simple — it includes customizations, sizes, combos, limited-time items, and regional variations. Training an AI to reliably handle all of this, for all customers, in all conditions, is an enormous technical challenge that was not fully solved.

3. Poor Failure Handling

When the AI did not understand an order, its responses were often unhelpful or made things worse (like adding 69 packets of butter). Good AI systems need to handle uncertainty gracefully — asking for clarification in a natural way, or handing off to a human smoothly, rather than guessing incorrectly. This aspect of the system was poorly designed.

4. Customer Experience Was Not Prioritized

Technology decisions should always center on the end user. Drive-thru customers care about **speed, accuracy, and ease**. The AOT system delivered none of these reliably. Customers had to repeat themselves, correct errors, and wait while the system struggled — the exact opposite of the intended experience.

A critical question that should be asked before deploying any customer-facing technology is: **What happens when it goes wrong, and how does that affect the customer’s experience of our brand?**

5. Viral Social Media Risk

In the era of TikTok and Twitter/X, a flawed customer experience is not just a customer service problem — it is a public relations problem. The videos of McDonald’s AI ordering outrageous quantities of butter were funny, but they were also damaging. They shaped public perception of the technology and put pressure on the company to act.

What McDonald’s Learned (And What It Should Have Done)

McDonald’s has said it continues to explore AI in other areas of its operations — kitchen management, inventory forecasting, digital menu personalization — where AI is less visible to customers and where errors have lower stakes.

Experts suggest that a better approach to the drive-thru AI might have included:

- **Longer, more rigorous pilot testing** in diverse real-world environments before broad deployment
- **A clearer human handoff protocol** — when the AI is uncertain, it should immediately route to a human rather than guessing
- **Narrower initial scope** — perhaps using AI only for simple, common orders before tackling complex customizations
- **Genuine measurement of success** — defining upfront what “good enough” looks like and stopping if those benchmarks are not met

The Bigger Picture: AI Hype vs. AI Reality

The McDonald’s case is part of a broader pattern across industries. Companies feel competitive pressure to adopt AI quickly, often before the technology is mature enough for the specific task. This is sometimes called “**AI washing**” — deploying AI tools to appear innovative rather than because they genuinely solve a problem better than existing approaches.

This does not mean AI has no place in fast food or customer service. It means that deploying AI well requires honest assessment of the technology’s limitations, rigorous testing, and a clear-eyed view of what success looks like.

Key Concepts for Discussion

Concept	Definition	Relevance to This Case
Natural Language Processing (NLP)	AI technology for understanding human speech/text	Core technology behind the voice ordering system
Pilot Testing	Testing a system at small scale before full deployment	Piloting at 100 locations was not enough to catch real-world failures
Production Gap	Difference in performance between testing and real-world use	The AI worked in the lab but not in the noisy drive-thru
Customer Experience (CX)	The overall experience a customer has interacting with a company	The AI degraded the drive-thru experience rather than improving it
Technology Governance	Oversight processes for deciding when and how to deploy technology	McDonald's lacked clear criteria for when to stop the program

Analysis Questions

1. Describe the company featured in the case, including its competitive advantage (or “moat”), historical background, and the nature of its industry or business operations.
2. Summarize the key facts presented in the case. Do these facts relate to or mirror any other cases discussed in class? If so, what insights or lessons from those previous cases can be applied here?
3. Analyze the economic consequences of the information system failure described. If a similar failure occurred in today’s technological environment, would the impact be greater or smaller? How likely is such an incident to occur now?
4. Discuss the broader implications of the risks identified in the case for contemporary IT infrastructures, cloud computing services, and emerging artificial intelligence systems.
5. Identify the key human decision point in this case — the moment where a different choice by a person or team would most likely have prevented or significantly reduced the harm. Who was responsible, and what should they have done differently?
6. Identify and explain five key lessons or takeaways derived from this case study.

Discussion Questions

1. What does this case reveal about where organizations tend to underestimate risk — in their technology, their processes, or their people? Use specific evidence from the case to support your answer.
2. If you were the CIO of the organization in this case, what are the first three changes you would make after this incident, and why?
3. McDonald’s acquired Apprento, then sold the technology to IBM, then ended the IBM partnership entirely. What does this build-vs-buy-vs-partner arc tell us about how organizations should govern technology decisions over time? Who should have the authority to end a failing technology program?

4. At what point should McDonald's have recognized the experiment was failing and pulled the plug? What metrics or signals should they have been tracking from the start?
 5. McDonald's is a data-rich company with massive resources. If they struggled to implement AI voice ordering, what does that suggest for smaller organizations considering similar technologies?
 6. Are there applications of AI within McDonald's operations — not customer-facing — where the technology might have a better chance of success? What makes those contexts different?
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Suggested Further Reading

- IBM and McDonald's press releases on the AOT partnership ([2021](#)) and termination ([2024](#))
 - "McDonald's to end AI drive-thru test with IBM" — [CNBC](#), 2024
 - Overview of natural language processing for business applications
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Instructions

1. Use **Bloomberg News** as the primary source for case research. You may also reference other *reputable* public websites or databases. All sources must be properly cited in **APA format** within your written report.
 2. The **written case report** should be between **5 and 10 double-sided pages**, inclusive of tables, figures, infographics, or any analytical materials. Acceptable submission formats include **Canva, Microsoft Word, PDF, Markdown, or LaTeX**.
 3. The **case presentation** should consist of **10 to 30 slides**, incorporating tables, charts, infographics, or analytical visuals as needed. Acceptable file formats include **Canva, Microsoft PowerPoint, PDF, Quarto Markdown, or LaTeX**.
 4. Include a **detailed log of each team member's contributions** as an appendix to the written report. Every team member must be prepared to address questions during the presentation, both on the overall case content and specifically on the sections they contributed to.
 5. **AI tools may not be used** to generate written content for the report. Collaborative work through **Google Docs** (with version control and simultaneous editing) is encouraged to demonstrate authentic team contribution and transparency in authorship. All version control tracking links must be shared with the instructor when asked.
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Prepared for Suresh Paul's MIS Course — Case Study Student Reading. Not intended for commercial use. Not intended for AI training